MINIMAL PRIMES OF THE SUPPORT, OF THE ASSOCIATED PRIMES AND OF A FILTRATION ALL AGREE

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Allowed standard prerequisites (editorial).

- Associated primes of a filtration with cyclic prime quotients. Invoked in the counted proof at the start (lines 15419-15421). Identifies the minimal primes of the support with those of the primes of a filtration.
- Minimality of a prime in the support versus in the associated primes. Used in the second half of the counted proof (lines 15423-15447). Transfers minimality between the two sets of primes through annihilators.

Source reference. Original lemma 0586: sources/stacks-algebra/pinned-original-source.tex, source lines 15335–15335. Navigation pointer for \ref{lemma-ass-support} used inside the counted span (line 15436); the argument that consumes it is retained in the span.

Source reference. Original lemma 00L7: sources/stacks-algebra/pinned-original-source.tex, source lines 15234–15234. Navigation pointer for \ref{lemma-filter-minimal-primes-in-support} used inside the counted span (line 15420); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proposition 0.1. *Let R be a Noetherian ring. Let M be a finite R-module. The following sets of primes are the same:*

(1) *The minimal primes in the support of M.*
(2) *The minimal primes in $Ass(M)$.*
(3) *For any filtration $0 = M_0 \subset M_1 \subset \ldots \subset M_{n-1} \subset M_n = M$ with $M_i/M_{i-1} \cong R/\mathfrak{p}_i$ the minimal primes of the set $\{\mathfrak{p}_i\}$.*

Proof. Choose a filtration as in (3). In Lemma 00L7 we have seen that the sets in (1) and (3) are equal.

Let $\mathfrak{p}$ be a minimal element of the set $\{\mathfrak{p}_i\}$. Let i be minimal such that $\mathfrak{p} = \mathfrak{p}_i$. Pick $m \in M_i$, $m \notin M_{i-1}$. The annihilator of m is contained in $\mathfrak{p}_i = \mathfrak{p}$ and contains $\mathfrak{p}_1\mathfrak{p}_2 \ldots \mathfrak{p}_i$. By our choice of i and $\mathfrak{p}$ we have $\mathfrak{p}_j \not\subset \mathfrak{p}$ for $j < i$ and hence we have $\mathfrak{p}_1\mathfrak{p}_2 \ldots \mathfrak{p}_{i-1} \not\subset \mathfrak{p}_i$. Pick $f \in \mathfrak{p}_1\mathfrak{p}_2 \ldots \mathfrak{p}_{i-1}$, $f \notin \mathfrak{p}$. Then fm has annihilator $\mathfrak{p}$. In this way we see that $\mathfrak{p}$ is an associated prime of M. By Lemma 0586 we have $\mathrm{Ass}(M) \subset \mathrm{Supp}(M)$ and hence $\mathfrak{p}$ is minimal in $\mathrm{Ass}(M)$. Thus the set of primes in (1) is contained in the set of primes of (2).

Let $\mathfrak{p}$ be a minimal element of $\mathrm{Ass}(M)$. Since $\mathrm{Ass}(M) \subset \mathrm{Supp}(M)$ there is a minimal element $\mathfrak{q}$ of $\mathrm{Supp}(M)$ with $\mathfrak{q} \subset \mathfrak{p}$. We have just shown that $\mathfrak{q} \in \mathrm{Ass}(M)$. Hence $\mathfrak{q} = \mathfrak{p}$ by minimality of $\mathfrak{p}$. Thus the set of primes in (2) is contained in the set of primes of (1). □